DiLAST: Leveraging Differential RGB Features for Hyperspectral Image Super-Resolution

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***Abstract*—Hyperspectral image super-resolution aims to reconstruct high-quality spatial–spectral cubes from RGB images. However, the limited spectral coverage of RGB inputs hinders the simultaneous modeling of spatial structures and spectral relationships. To address this limitation, we propose a differential low-rank adaptive spatial–spectral Transformer (DiLAST). Initially, a differential operator is employed to enhance RGB features in a bottom–up manner, explicitly amplifying subtle interchannel differences. The enhanced features are, then, fed into a U-shaped backbone, which integrates three complementary modules for joint spatial–spectral modeling. Specifically, a center spatial–spectral attention (CSSA) module employs cross-attention mechanisms to capture local-to-global dependencies across both spatial and spectral domains; an adaptive cross-scale fusion (ACF) module utilizes learnable gating weights to establish dynamic interaction pathways between shallow high-frequency details and deep semantic representations; and a low-rank spectral calibration (LRSC) module exploits low-rank matrix priors to reveal low-dimensional manifold structures among spectral bands, thereby enhancing spectral consistency. By leveraging the synergistic effects of spatial nonlocality, global spectral correlation, and low-rank properties, the proposed DiLAST achieves peak signal-to-noise ratio (PSNR) improvements of 33.82, 36.03, and 37.01 dB on benchmark datasets. Moreover, the accuracy and practical applicability of the reconstructed spectra have been effectively validated in remote sensing scenarios and object tracking tasks. The code is accessible at https://github.com/renqi1998/DiLAST**

***Index Terms*—Adaptive cross-scale fusion (ACF), center spatial–spectral attention (CSSA), hyperspectral image (HSI), low rank, spectral super-resolution (SSR), Transformer.**

I. Introduction

HYPERSPECTRAL image (HSI) significantly overcomes the spectral limitations of RGB and multispectral imaging by leveraging spaceborne or airborne spectrometers that capture radiation across hundreds of contiguous spectral bands with sub-nanometer resolution. The resulting dense spectral sampling facilitates the identification of unique material signatures at specific wavelengths, thereby providing rich spectral and spatial information. Therefore, HSI plays a crucial role in remote sensing tasks such as landcover classification [1], target tracking [2], and super-resolution reconstruction [3].

Hyperspectral imaging faces intertwined challenges in hardware, theory, and algorithms. Complex and costly push-broom or array detectors limit real-time capability [4], while the Shannon–Nyquist sampling theorem imposes a spatial–spectral tradeoff and spectral mixing further complicates reconstruction [5]; moreover, conventional methods based on low-rank manifold assumptions fail to recover fine spectral details in high-dimensional spaces. Among two main research trajectories for addressing these challenges, multisource fusion exploits spectral–spatial complementarity via multisource synergy but is constrained by stringent spatiotemporal consistency across heterogeneous sensors, whereas spectral super-resolution (SSR) offers a more flexible solution by learning nonlinear end-to-end mappings to reconstruct high-dimensional spectral cubes directly from RGB images, thus eliminating reliance on auxiliary sensors [3], [6], [7], [8], [9], [10]. Traditional SSR methods predominantly adopt a model-driven paradigm, enhancing spectral resolution by leveraging the inherent correlations between hyperspectral and RGB observations. These approaches are typically supported by handcrafted priors that encode domain-specific knowledge. Sparse coding, as a representative method, employs sparse representation techniques to construct an overcomplete dictionary that spans all hyperspectral pixels [11], [12], [13], [14]. This enables effective reconstruction of high-resolution spectra while potentially reducing acquisition costs. To further preserve spatial coherence, spatial constraints are integrated into the reconstruction process—either by enforcing local consistency among neighboring pixels or through the unmixing of abundance maps in the reconstructed HSI [15], [16], [17]. However, SSR remains a fundamentally ill-posed inverse problem. Although model-based techniques are intuitive and computationally efficient, their reliance on task-specific handcrafted priors inherently limits their generalization and expressive power, often resulting in degraded spectral fidelity under complex scenarios, diverse sensing conditions, or real-world variations. These limitations underscore the need for data-driven approaches capable of learning robust spectral–spatial

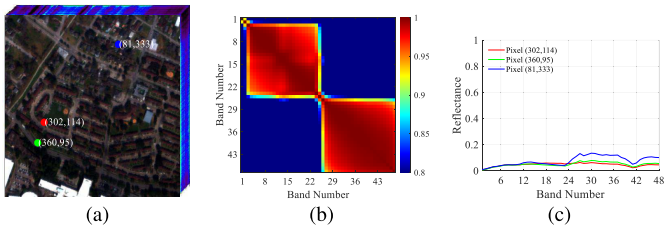
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(a) (b) (c)

Fig. 1. (a) Example HSI from the DFC2018 Houston dataset with sampled pixel locations. (b) Bandwise CC matrix. (c) Spectral curves at selected pixels reveal intraclass variability within vegetation and unexpected interclass similarity between vegetation and water in specific bands.

representations directly from observations. The advancement of deep learning in recent years has significantly propelled research in the domain of SSR. To accelerate research in this field, the New Trends in Image Restoration and Enhancement (NTIRE) challenges held in 2018 and 2020 [18], [19] spurred the development of numerous convolutional neural network (CNN)-based SSR frameworks. These methods improved reconstruction quality by learning end-to-end nonlinear mappings from RGB images to hyperspectral representations [20], [21], [22], [23], [24]. Despite their effectiveness, CNN-based models are inherently limited in capturing long-range spatial dependencies and exploiting nonlocal spectral correlations due to their localized receptive fields. To address these limitations, Transformer-based architectures have been introduced into SSR. By leveraging self-attention mechanisms, they effectively model global contextual relationships and long-range interactions across both spatial and spectral dimensions, leading to substantial improvements in spectral reconstruction accuracy and generalization [9], [25], [26], [27], [28].

Beyond capturing global context, accurately modeling spectral correlations is essential in HSI-SSR, as it directly impacts the reconstruction of spectra along the spectral dimension spectra along the spectral dimension. As illustrated in Fig. 1, we take the DFC2018 Houston dataset [29] as an example. The red and green markers denote two vegetation pixels exhibiting notable intraclass spectral variability. Meanwhile, the blue marker corresponding to water shows spectral similarity to the red pixel in certain bands, reflecting interclass overlap. These observations highlight the necessity for SSR models to capture both shared and discriminative spectral characteristics to ensure accurate and physically meaningful HSI reconstruction. However, existing Transformer-based SSR models face three significant challenges in addressing these spectral correlations. First, most existing methods emphasize spectral attention while underutilizing spatial dependencies, resulting in insufficient spatial context modeling. Moreover, comprehensive global spatial modeling often incurs significant computational overhead [9], [10]. Second, the prevailing spectral self-attention mechanisms often learn full-rank correlation matrices and fail to adequately exploit the low-rank nature of HSI even though such data typically reside in low-rank subspaces that contain critical feature information [25], [30], [31]. Lastly, common strategies like feature addition or concatenation are insufficient for addressing differences in feature importance between the encoder and decoder stages. This limitation restricts the effective differentiation among various feature channels and spectral bands [9], [10], [32], [33], [34]. Collectively, these challenges constrain both the reconstruction accuracy and generalization ability of SSR models. They underscore the urgent need for refined spatial–spectral joint modeling frameworks that incorporate low-rank priors and adopt adaptive, structure-aware feature fusion mechanisms.

To address the limitations of existing reconstruction methods, this article proposes a differential low-rank adaptive spatial–spectral Transformer (DiLAST), as illustrated in Fig. 2, which jointly exploits nonlocal spatial similarity and global spectral correlation of HSI with minimal computational overhead. Specifically, a differential operator is first applied to amplify subtle differences among the RGB channels, yielding more discriminative initial features. A center spatial–spectral attention (CSSA) module is subsequently introduced. The center window self-attention (CWSA) component captures nonlocal spatial similarities within local center windows, while the global spectral self-attention (GSSA) explicitly models the highly correlated bands along the spectral dimension. Next, an adaptive cross-scale fusion (ACF) module dynamically weights shallow and deep features to emphasize detail-rich regions and enable efficient multiscale information fusion. Finally, a low-rank spectral calibration (LRSC) module employs a global spectral memory unit to project the spatial–spectral cube into a low-rank vector space, aggregating the most representative spectral statistics and enhancing spectral consistency. By synergistically leveraging spatial nonlocality, spectral correlation, and low-rank priors, DiLAST achieves superior performance compared to state-of-the-art (SOTA) methods on benchmark datasets and real-world scenarios, while also demonstrating generalization in downstream tasks.

This article's key contributions are summarized as follows.

1) We propose a new framework, DiLAST, which first enhances RGB representations via differential operators and then integrates them into a Transformer-based architecture to concurrently and explicitly model spectral similarity and disparity.
2) To incorporate a spatial–spectral coupling prior, we devise a trainable CSSA module that aggregates interactions between nonlocal spatial self-similarity and global spectral correlation during HSI reconstruction. Acting in a spatial-spectrally consistent self-attention scheme, CSSA enables collaborative learning and effectively drives joint spatial–spectral optimization of the estimated HSI within the network.
3) We incorporate the ACF module to adaptively weight shallow and deep features, enabling cross-scale Transformer interactions. In parallel, the LRSC module uses a global memory vector to aggregate low-rank spectral patterns, enhancing spectral consistency and robustness in spatial–spectral representation.
4) The experimental results demonstrate that DiLAST consistently outperforms SOTA SSR methods on both benchmark datasets and real-world scenarios, achieving superior visual quality and quantitative performance. Furthermore, its effectiveness and practicality are

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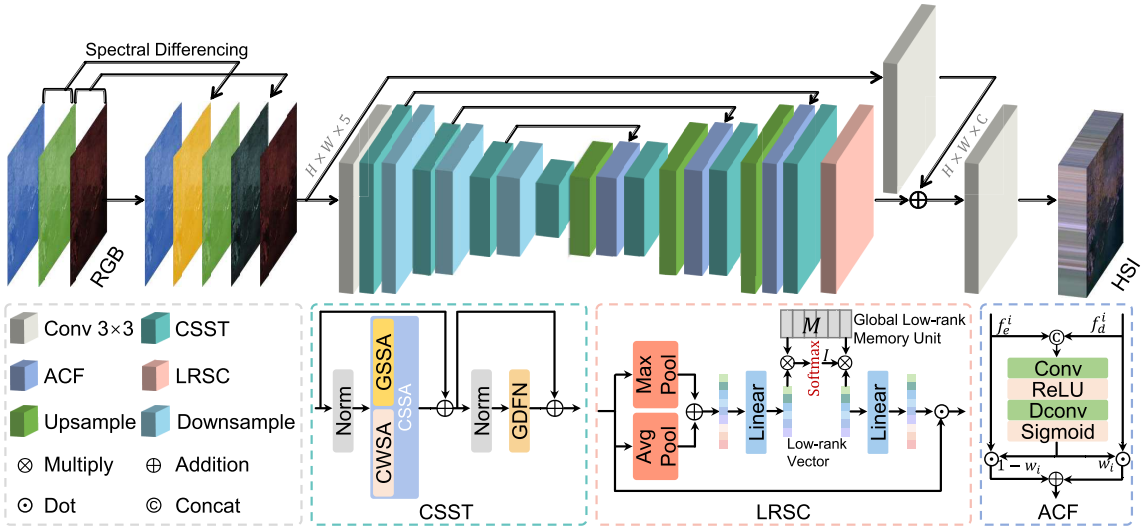


Fig. 2. Schematic illustration of the proposed DiLAST framework for HSI-SSR is shown. CSST refers to the center spatial–spectral Transformer, ACF denotes the adaptive cross-scale fusion module, and LRSC represents the low-rank spectral calibration module.

validated through improved performance in object tracking tasks.

The remainder of this article is organized as follows. Section II reviews relevant works on HSI-SSR. Section III details the proposed framework and its core components. The experimental results and analyses are presented in Section IV, followed by conclusions and future directions in Section VI.

II. Related Work

A. Hyperspectral Image Acquisition

Traditional hyperspectral imaging systems acquire spatial and spectral information either through spatial scanning, which sequentially captures spatial data over time, or spectral scanning, which entails sequentially capturing images at different wavelengths [35], [36]. However, inherent hardware limitations make the acquisition of high-resolution HSI time-consuming, rendering them impractical for dynamic or real-time imaging scenarios. To address this challenge, snapshot compressive imaging (SCI) systems have been developed, which capture spectral data in a single exposure using a detector array and subsequently reconstruct the HSI [35], [37], [38]. Among these, coded aperture snapshot spectral imaging (CASSI), based on compressive sensing theory, has garnered significant attention for its ability to recover high-dimensional spectral information from limited measurements [39], [40]. Despite these advancements, the high implementation costs associated with SCI systems impede their widespread adoption in practical applications. In contrast, RGB images are inexpensive and easily accessible, necessitating only commercially available devices such as smartphones or consumer-grade cameras. Reliable HSI reconstruction from RGB images can significantly reduce acquisition costs and improve efficiency, making SSR a promising and cost-effective alternative to traditional HSI systems that is attracting growing research interest.

B. Spectral Reconstruction From RGB Images

The high cost and complexity of hyperspectral imaging devices have made reconstructing HSIs from RGB images a major research focus in recent years. Existing SSR methods can be broadly classified into two primary categories: physics-driven approaches [12], [14], [26], [33], which leverage prior knowledge and physical models to map RGB images to their hyperspectral counterparts and learning-driven methods [9], [10], [25], [41], which exploit datasets to learn nonlinear mappings without relying on explicit physical modeling.

Physics-driven approaches rely on traditional techniques such as sparse dictionary learning and tensor decomposition. Parmar et al. [42] incorporated sparse priors to jointly reconstruct spectral cubes by integrating spatial–spectral information and sparsity constraints. Arad and Ben-Shahar [43] developed a sparse dictionary to model the relationship between RGB images and their corresponding HSI, enabling effective reconstruction. Timofte et al. [44] introduced a fast super-resolution method combining dictionary features with anchored regressors, while Akhtar et al. [45] advanced a nonparametric Bayesian sparse representation approach that synthesizes low- and high-resolution images to reconstruct HSIs without dedicated hyperspectral imaging equipment. Aeschbacher et al. [46] added low-rank constraints into dictionary learning to maintain global spectral consistency and reduce spectral distortion. Fotiadou et al. [11] proposed a coupled dictionary learning scheme in a shared feature space, whereas Yi et al. [12] enforced separate constraints on spatial and spectral components. Gao et al. [14] exploited overlapping regions between hyperspectral and multispectral images to learn low-rank dictionaries, and Hong et al. [47] incorporated local consistency constraints to guide reconstruction. Despite their effectiveness, these methods rely on hand-crafted priors and domain-specific assumptions, which limit generalization and degrade spectral fidelity in diverse conditions, thereby motivating the development of learning-based approaches that address these issues through data-driven modeling.

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Building on their success in natural image processing, CNNs have been widely adopted in remote sensing for SSR to learn nonlinear mappings from RGB images to hyperspectral representations. Stiebel et al. [48] first implemented spectral reconstruction from RGB inputs using a U-Net architecture, establishing the foundation for subsequent CNN-based SSR methods. Xiong et al. [49] later introduced the hyperspectral super-resolution CNN (HSCNN), based on very deep super-resolution frameworks, inspiring enhanced variants such as HSCNN+ [20], which ranked first in the NTIRE 2018 challenge. In the NTIRE 2020 competition, Arad et al [19]. proposed the adaptive weighted attention network (AWAN), which integrated dual residual attention mechanisms to further improve reconstruction. Zhao et al. [23] developed a hierarchical regression network (HRNet) that combined residual modules with attention mechanisms to mitigate artifacts and noise, while Li et al. [32] incorporated residual connections into an encoder–decoder framework, achieving third place in NTIRE 2022 challenge. Wu et al. [50] proposed a lightweight spatial–spectral module designed to improve reconstruction accuracy with reduced computational overhead. Similarly, Hu et al. [51] introduced HDNet, a dual-domain network with spatial–spectral attention, to better preserve high-resolution information. CNN-based SSR methods address the ill-posed RGB-to-HSI mapping by learning spectral priors from large datasets, enhancing reconstruction fidelity.

Due to their limited receptive fields, CNNs struggle with long-range spatial–spectral modeling, which has prompted interest in Transformer architectures. Cai et al. [9] proposed MST++, a stacked Transformer-based framework that applies multihead self-attention (MSA) in the spectral domain to effectively model global spectral correlations. Building upon MST++, Wang et al. [25] introduced the hybrid spectral attention Transformer (HySAT), which jointly models spectral similarity and distinctiveness to address MST++'s limitations in representing interband variations. Yang et al. [10] further explored spectral–spatial fusion by extracting complementary structural information from RGB channels, while Du et al. [27] proposed a hybrid network that combines 3-D spatial–spectral convolution with a U-shaped spectral Transformer, to integrate shallow and deep representations. While these methods improve either spectral or spatial modeling, most focus on a single domain and lack a unified spatial–spectral representation framework.

Related work in spatial super-resolution has also influenced SSR research. FM-Net [52] is inspired by micelle aggregation to enhance texture, HTINet [53] by heat diffusion principles for smooth propagation, and CCN [54] imposes curvature consistency for geometric fidelity. ESSAformer [55] employs a Transformer with spectral correlation coefficient (SCC)-based attention for spatial HSI enhancement but assumes complete spectral input. While effective in spatial enhancement tasks, these methods do not address the spectral reconstruction problem from RGB inputs.

Despite progress, existing SSR methods struggle to jointly model spatial–spectral dependencies, preserve spectral fidelity under diverse conditions, and generalize across datasets. To address these challenges, we propose the DiLAST framework that captures nonlocal spatial self-similarity, global spectral correlation, and low-rank priors to enhance reconstruction accuracy and spectral consistency.

III. Proposed Method

A. Framework Overview

In RGB-to-HSI imaging, the HSI is typically represented as $\mathbf{Y} \in \mathbb{R}^{H\times W\times C}$, where H, W, and C denote the image height, width, and the number of spectral bands, respectively. Its degradation to an RGB image $\mathbf{X} \in \mathbb{R}^{H\times W\times 3}$ through the spectral response of a camera can be modeled as

$$\mathbf{X}^{(c)}(x, y, :) = \boldsymbol{\phi}\,\mathbf{Y}(x, y, :), \quad \boldsymbol{\phi} \in \mathbb{R}^{3\times C} \tag{1}$$

where $\boldsymbol{\phi}$ denotes the spectral response matrix. This process constitutes an injective degradation that compresses high-dimensional spectral content into only three channels, often introducing irreversible distortions [8].

To mitigate such degradation, DiLAST first applies *spectral differencing* to the RGB image $\mathbf{X}$ by computing the channel-wise differences

$$\mathbf{G}^{(k)}(x, y) = \mathbf{X}^{(k)}(x, y) - \mathbf{X}^{(k+1)}(x, y), \quad k = 1, 2. \tag{2}$$

The original RGB channels are concatenated with the differential maps to form an enhanced input cube

$$\mathbf{X}_{\text{enh}} = [\mathbf{X}, \mathbf{G}] \in \mathbb{R}^{H\times W\times 5} \tag{3}$$

which amplifies subtle interchannel differences and strengthens the spectral representation at the input level.

The enhanced input $\mathbf{X}_{\text{enh}}$ is projected to a higher dimensional feature space through a 3×3 convolution and passed into a multiscale U-shaped backbone (Fig. 2). This backbone integrates hierarchical downsampling and upsampling to model both fine spatial details and long-range context while maintaining computational efficiency. The deep features are subsequently processed by three specialized modules: CSSA, ACF, and LRSC, which collaboratively reconstruct spatial–spectral information.

The encoder downsamples spatial resolution via 4×4 stride-2 convolutions, doubling channels at each stage, while the decoder symmetrically upsamples using 2×2 stride-2 transposed convolutions. Each stage embeds stacked center spatial–spectral Transformer (CSST) blocks to capture nonlocal spatial similarity and global spectral correlations. Skip connections preserve structural details, and a long-range residual path enhances feature coherence and stabilizes training.

To enhance interstage information flow, we introduce an ACF module that adaptively reweights features across different depths, and an LRSC module that refines spectral representations. In synergy, these modules improve both reconstruction accuracy and robustness. Sections III-B–III-D detail the design and implementation of the CSSA, ACF, and LRSC modules.

B. Center Spatial–Spectral Transformer

HSI is typically represented as 3-D cubes with strong spatial and spectral similarities. Effectively leveraging these inherent similarities is critical for salient features extraction

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and accurate spectral reconstruction. However, most deep learning-based HSI reconstruction methods treat spatial and spectral dependencies independently, lacking a unified framework for joint spatial–spectral learning. CNN-based approaches primarily capture spatially invariant local features through convolutions with limited receptive fields, making it difficult to model nonlocal similarities [32], [50], [56]. In contrast, Transformer-based methods excel in modeling global spectral dependencies through self-attention mechanisms, yet often underperform in capturing fine-grained spatial details [9], [10], [25], [27]. Consequently, effectively modeling spatial and spectral similarities remains a significant challenge in advancing SSR. To address this, we propose the CSST with a multihead spatial–spectral attention mechanism that jointly exploits nonlocal spatial similarity and strong spectral correlation. At its core lies the CSSA module, which captures both spatial and spectral dependencies in a unified manner.

To enhance stability and prevent rank collapse, CSST incorporates architectural components such as skip connections, layer normalization (LN), and a multilayer perceptron (MLP). Given input features $F_{l-1} \in \mathbb{R}^{H\times W\times C}$ at the lth layer, the CSST module is defined as

$$\begin{cases} F_l' = \mathrm{CSSA}\left(\mathrm{LN}\left(F_{l-1}\right)\right) + F_{l-1} \\ F_l = \mathrm{MLP}\left(\mathrm{LN}\left(F_l'\right)\right) + F_l' \end{cases} \tag{4}$$

where F_l' is the intermediate feature after attention, and F_l is the final output after transformation.

The internal architecture of the CSSA, as depicted in Fig. 2, comprises a CWSA layer and a GSSA layer. The CWSA layer is designed to model local spatial dependencies, while the GSSA layer addresses global spectral relationships. The attention outputs are computed as

$$\begin{cases} F_i^{\mathrm{ns}} = \mathrm{CWSA}\left(F^{\mathrm{in}}\right) \\ F_i^{\mathrm{gs}} = \mathrm{GSSA}\left(F^{\mathrm{in}}\right). \end{cases} \tag{5}$$

These two attention outputs are fused via a learnable scalar λ to form the final CSSA output

$$\underset{i}{\overset{CSSA}{F}} = \mathrm{CSSA}\left(\underset{i}{\overset{\mathrm{gs}}{F}} \cdot \lambda + \underset{i}{\overset{\mathrm{ns}}{F}}\right) \tag{6}$$

where F^{in} denotes the normalized input, F_i^{ns} and F_i^{gs} are the outputs of CWSA and GSSA, respectively, and λ balances the contributions of spatial and spectral representations.

1) Center Window Self-Attention: To effectively model local spatial dependencies while reducing computational overhead, we propose a CWSA mechanism, as shown in Fig. 3. While global spatial self-attention is effective at capturing long-range dependencies, its high computational cost poses challenges for HSI-SSR applications. To alleviate the computational burden of global self-attention, approaches such as the Swin Transformer [57] and CSWin Transformer [58] restrict the attention scope via local- or cross-window strategies. However, as illustrated by the similarity heatmaps in Fig. 4, center pixels generally exhibit stronger correlations with nearby pixels, while similarities decrease with increasing spatial distance. This observation underscores the importance of capturing locally dense spectral–spatial features in HSI,

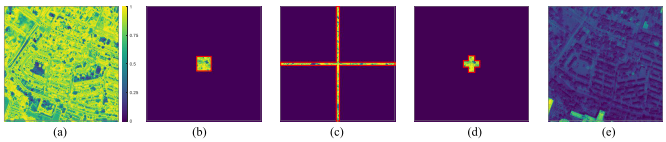


Fig. 3. Similarity statistics from the DFC2018 Houston dataset, showing reduced similarity with increasing spatial distance. (a) Original feature map. (b) Window self-attention [57] captures local similarity in small neighborhoods. (c) Cross-shaped self-attention [58] models long-range dependencies along horizontal and vertical axes. (d) CWSA (ours) combines local and cross patterns for enhanced contextual aggregation. (e) GSSA (ours) highlights widespread high similarity across distant regions, indicating strong global spectral correlations. The color bar denotes normalized spectral similarity, with yellow indicating stronger and purple indicating weaker similarity between spectral feature vectors.

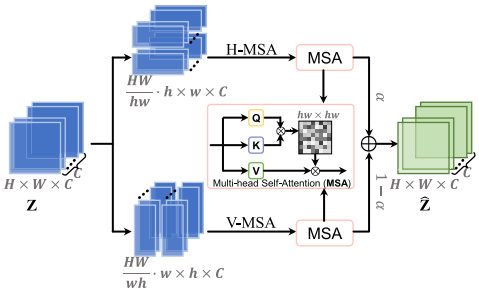


Fig. 4. Schematic representation of the proposed CWSA.

while also motivating attention mechanisms that balance local context modeling with the ability to capture long-range dependencies. Although the CSWin Transformer integrates cross-window attention, it often involves interactions with less informative regions, limiting its ability to capture essential local structures. Our CWSA attends to information-rich central neighborhoods with partial nonlocal context, balancing precision and efficiency. To adapt to varying receptive fields, we employ multishape windowing with horizontal and vertical center windows across stages.

The input feature $F \in \mathbb{R}^{H\times W\times C}$ is defined and performs window-based self-attention in both horizontal and vertical directions. These operations are fused via a learnable parameter α, yielding the output

$$\hat{F} = \alpha \cdot \text{H-MSA}\left(F\right) + (1-\alpha)\cdot \text{V-MSA}\left(F\right). \tag{7}$$

Here, H-MSA and V-MSA refer to self-attention within horizontal and vertical window configurations, respectively.

For horizontal window of size $[h, w]$, where $h > w$, we divide the feature map into $N = (H \times W/h \times w)$ nonoverlapping windows, represented as $\{F_1, F_2, \ldots, F_N\}$, each of shape $\mathbb{R}^{h\times w\times C}$. Within each subwindow F_i, we compute query $\mathbf{Q}_i$, key $\mathbf{K}_i$, and value $\mathbf{V}_i$ as

$$\mathbf{Q}_i = \boldsymbol{W}_d^{\mathbf{Q}}\boldsymbol{W}_p^{\mathbf{Q}}F_i, \quad \mathbf{K}_i = \boldsymbol{W}_d^{\mathbf{K}}\boldsymbol{W}_p^{\mathbf{K}}F_i, \quad \mathbf{V}_i = \boldsymbol{W}_d^{\mathbf{V}}\boldsymbol{W}_p^{\mathbf{V}}F_i \tag{8}$$

where $\boldsymbol{W}_d^{(\cdot)}$ and $\boldsymbol{W}_p^{(\cdot)}$ represent the 1×1 pointwise convolution and 3×3 depthwise convolution, respectively. Each feature is reshaped $\mathbb{R}^{h\times w\times C}$ to $\mathbb{R}^{hw\times C}$ before attention computation.

The window-level attention output is computed as

$$\hat{F}_i = \mathrm{Softmax}\left(\frac{\mathbf{Q}_i\mathbf{K}_i^{\mathrm{T}}}{\sqrt{d}} + P\right)\mathbf{V}_i \tag{9}$$

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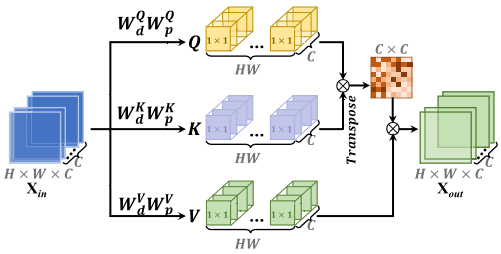


Fig. 5. Schematic of the proposed GSSA.

where P is a learnable positional embedding, and d signifies the feature dimension. After processing all windows, we reconstruct the full feature map via

$$\text{H-MSA}(F) = \text{Merge}\left(\hat{F}_1, \hat{F}_2, \ldots, \hat{F}_N\right) \tag{10}$$

where Merge denotes reversing the window partition to recover the original spatial dimensions. The V-MSA uses vertical windows of size $[w, h]$ (with $w > h$) and follows the same procedure.

2) Global Spectral Self-Attention: While the proposed CWSA module effectively captures local spatial dependencies, it provides limited capacity in modeling spectral correlations, which are crucial for HSI tasks. HSI exhibits strong interband correlations, which serve as valuable priors for spectral attention, as shown in Fig. 1(b). However, as shown in Fig. 3, the original feature map in Fig. 3(a) primarily captures local correlations, whereas Fig. 3(e) reveals that GSSA enhances global similarity across distant regions. To leverage this, we introduce a GSSA module [59], as depicted in Fig. 5, which learns a global attention map among spectral channels, thereby improving the discrimination and expressive power of spectral features and explicitly preserving global spectral consistency, as demonstrated in Fig. 3(e).

Given input $F \in \mathbb{R}^{H\times W\times C}$, we first compute the query $\mathbf{Q}$, key $\mathbf{K}$, and value $\mathbf{V}$ representations using 1×1 pointwise and 3×3 depthwise convolutions

$$\mathbf{Q} = \boldsymbol{W}_d^{\mathbf{Q}}\boldsymbol{W}_p^{\mathbf{Q}}F, \quad \mathbf{K} = \boldsymbol{W}_d^{\mathbf{K}}\boldsymbol{W}_p^{\mathbf{K}}F, \quad \mathbf{V} = \boldsymbol{W}_d^{\mathbf{V}}\boldsymbol{W}_p^{\mathbf{V}}F. \tag{11}$$

To compute spectral attention, we first aggregate spatial information by reshaping $\mathbf{Q}$, $\mathbf{K} \in \mathbb{R}^{H\times W\times C}$ to $\mathbb{R}^{C\times(HW)}$ and compute attention across channels

$$\hat{F} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathrm{T}}}{\sqrt{d}}\right)\mathbf{V} \tag{12}$$

where Softmax is applied along the channel dimension, and d denotes the feature dimension.

Following [9] and [59], we employ multihead channel attention, dividing the channel dimension into multiple groups and applying GSSA independently to each. This enables richer cross-channel modeling and improves the representation of complex spectral dependencies.

C. Adaptive Cross-Scale Fusion

Skip connections play a crucial role in U-Net architectures by restoring low-level details lost during downsampling [9], [34], [59]. These connections typically add or concatenate spatially aligned encoder–decoder features at corresponding resolutions. However, this implicitly assumes equal importance for shallow and deep features, disregarding differences in their information density. Such an imbalance across spectral bands and channels results in suboptimal feature fusion and hinders overall model performance.

To address this issue, we propose an ACF mechanism. As illustrated in Fig. 2, the ACF module decouples feature extraction and reweighting into two stages. It first estimates the importance of encoder and decoder features via convolutional attention, then explicitly reweights and fuses them. Let i denote the layer index. The process is defined as

$$F = \text{LeakyReLU}\left(\boldsymbol{W}_p\left(\left[F_d^i \,\|\, F_e^i\right]\right)\right) \tag{13}$$

$$M_i = \sigma\left(\boldsymbol{W}_d(F)\right) \tag{14}$$

$$F_d^{i+1} = \textbf{Conv}\underbrace{\left((1 - M_i)\odot F_d^i + M_i \odot F_e^i\right)}_{\text{feature reweighting}} \tag{15}$$

where σ denotes the Sigmoid function, and $\boldsymbol{W}_p$, $\boldsymbol{W}_d$, and **Conv** denote the convolution kernel. The ACF module enhances significant spatial locations and spectral bands through an elementwise gating mechanism. It offers two main advantages: first, it avoids redundant convolutions on concatenated features, reducing model complexity and focusing computation on essential information. Second, it explicitly reweights encoder and decoder features, emphasizing high-information regions to improve fusion efficiency.

D. Low-Rank Spectral Calibration

Although model-driven methods often process HSI by dividing them into smaller blocks, recent studies have demonstrated that HSI exhibits significant low-rank characteristics in tasks such as denoising [31], compressed sensing [60], and spectral unmixing [61]. This indicates that a considerable amount of redundant spectral information can be effectively compressed into a low-dimensional subspace. Motivated by Li et al. [31], we propose an LRSC module to enhance spectral feature representation through memory-based low-rank modeling.

Let $F \in \mathbb{R}^{H\times W\times C}$ be the coarse features from the input. To obtain a compact spectral representation, we project F into a low-rank vector $F_k \in \mathbb{R}^{1\times 1\times K}$ using global average pooling, max pooling, and a linear projection

$$F_k = \left(\text{AvgPool}(F) + \text{MaxPool}(F)\right)W_k, W_k \in \mathbb{R}^{C\times K}. \tag{16}$$

We introduce a memory module that stores global spectral prototypes in a learnable memory matrix $\mathbf{M} \in \mathbb{R}^{K\times B}$ to enrich the representation. The correlation between F_k and memory entries $\mathbf{M}$ is computed as

$$I = \text{Softmax}(F_k\mathbf{M}) \tag{17}$$

where Softmax is applied along the memory dimension B. The refined low-rank vector F_l is, then, obtained via

$$F_l = I\mathbf{M}. \tag{18}$$

Finally, to recalibrate the original features F, we project F_l back to the original channel dimension using a weight matrix $W_c \in \mathbb{R}^{C\times K}$, and apply channelwise modulation

$$Y = F \odot W_c F_l \tag{19}$$

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TABLE I
DESCRIPTION OF THREE BENCHMARK DATASETS

Parameter	CAVE	NTIRE2020 / NTIRE2022
Camera	Cooled CCD	Specim IQ mobile / Specim PS Kappa DX4
Illumination	CIE Standard Illumination D65	Natural sunlight
Spectral resolution	10 nm	10 nm
Storage accuracy	16-bit	12-bit

where $\odot$ denotes the elementwise multiplication. This operation adaptively enhances spectral features based on the low-rank prior encoded in the memory.

E. Hybrid Loss Function

Previous methods utilize the mean relative absolute error (MRAE), denoted as $\mathcal{L}_{MRAE}$ as the loss function to quantify pixelwise differences between the reconstructed results and the ground truth across all bands. It is defined as follows:

$$\mathcal{L}_{\text{MRAE}} = \frac{1}{C}\sum_{i=1}^{C}\frac{|Y_i - \widehat{Y}_i|}{Y_i} \tag{20}$$

where $Y \in \mathbb{R}^{HW\times C}$ denotes the ground-truth HSI, and $\hat{Y}$ is the reconstructed result, with C representing the number of bands.

As shown in Fig. 1(b), adjacent bands in HSI generally exhibit strong correlations. To encourage spectral smoothness and constrain interband variation in the reconstructed HSI, we introduce a spectral consistency loss $\mathcal{L}_{\text{sc}}$, defined as

$$\mathcal{L}_{\text{sc}} = \sum_{i=1}^{C-1}\left\|\widehat{Y}_i - \widehat{Y}_{i+1}\right\|_2. \tag{21}$$

In addition, to better preserve edge structures and reduce local discontinuities in the spatial domain, we introduce a spatial smoothness loss $\mathcal{L}_{\text{ss}}$, which enhances spatial smoothness and structural consistency. It is formulated as

$$\mathcal{L}_{\text{ss}} = \sum_{ij}\sqrt{\left\{\widehat{Y}_{i+1,j} - \widehat{Y}_{i,j}\right\}^2 + \left\{\widehat{Y}_{i,j+1} - \widehat{Y}_{i,j}\right\}^2}. \tag{22}$$

Finally, the total loss function $\mathcal{L}$ is defined as a weighted sum of the above components

$$\mathcal{L} = \mathcal{L}_{\text{MRAE}} + \alpha\mathcal{L}_{\text{sc}} + \beta\mathcal{L}_{\text{ss}} \tag{23}$$

where $\alpha = 0.5$ and $\beta = 0.2$ control the relative contributions of spectral and spatial constraints.

IV. EXPERIMENT AND ANALYSIS

This section presents systematic experiments on benchmark datasets to comprehensively evaluate the reconstruction performance of DiLAST. Ablation studies are also conducted to analyze the contribution of its core modules. In addition, we design downstream tasks including remote sensing image reconstruction and object tracking, demonstrating the potential of DiLAST to boost performance without relying on dedicated hyperspectral sensors.

A. Dataset Description

This study evaluates SSR performance on three benchmark datasets: CAVE [62], NTIRE2020 [19], and NTIRE2022 [63]. The key characteristics and parameter configurations of these datasets are summarized in Table I.

1) CAVE Dataset: This dataset was curated by Columbia University [64] using the Apogee Alta U260 hyperspectral camera. It comprises 33 real-world object scenes, including natural fruits, artificial fruits, human faces, and plastic items. Each image has a spatial resolution of 512 × 512 pixels, and the spectral data span the 400–700-nm range across 31 bands. For dataset partitioning, 28 images are used for training, while the remaining 5 are reserved for validation and testing.

2) NTIRE2020 Dataset: This study employs the "clean" track of the NTIRE2020 challenge as the data source for the spectral reconstruction task. The images within this track are devoid of noise interference and are accompanied by predefined parameters for the camera response function, thereby ensuring high data quality. Each image possesses a spatial resolution of 512 × 482 pixels, with a spectral range extending from 400 to 700 nm at fixed 10-nm intervals, resulting in a total of 31 spectral bands. The original dataset comprises 450 training images, 10 validation images, and 20 test images. However, several RGB images in the official test set lack corresponding HSI, making them unsuitable for evaluation. To ensure consistency and fairness, we adopt the validation set as the test set and randomly select ten images from the remaining training data as the new validation set, resulting in ten images each for validation and testing.

3) NTIRE2022 Dataset: This study employs the hyperspectral reconstruction dataset provided by the NTIRE 2022 challenge. The dataset was acquired using the PS Kappa DX4 hyperspectral camera across a variety of scenes and comprises 1000 sample images. Each HSI contains 31 spectral bands, covering a wavelength range from 400 to 700 nm, with a spatial resolution of 482 × 512 pixels. It is noteworthy that some RGB images in the official test set do not have corresponding HSI data. To maintain the rigor and comparability of the experiments, we designated the number of samples in the training, validation, and test sets as 890, 50, and 50, respectively.

B. Implementation Details

To enhance generalization, we normalize RGB images to [0, 1] and randomly crop paired 128 × 128 patches from the training data. Data augmentation includes random rotations (0°, 90°, 180°, and 270°) and horizontal/vertical flips. Following MST++ [9], we use a batch size of 20 and the Adam

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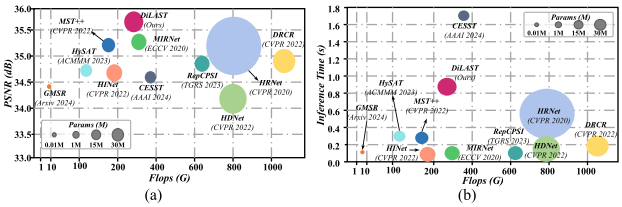


Fig. 6. Comparison of spectral reconstruction methods in terms of PSNR, FLOPs, parameters, and inference time. (a) PSNR versus FLOPs. (b) Inference time versus FLOPs. The radius of each circle denotes the number of parameters. DiLAST achieves a favorable balance between accuracy and efficiency, offering competitive PSNR with moderate computational cost, compact model size, and low inference time.

TABLE II
EFFECT OF SPECTRAL DIFFERENCING ON PERFORMANCE

Ablation Setting	RMSE	PSNR	SSIM	SAM	ERGAS	CC
RGB	4.9293	35.755	0.9833	6.01	18.4853	**0.9950**
Gaussian filtering	4.8627	35.983	0.9842	**5.62**	18.0471	0.9947
Spectral differencing	**4.7325**	**36.026**	**0.9849**	5.79	**17.8141**	0.9949

optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$, and weight decay of 1×10^{-6}) with an initial learning rate of 1×10^{-4} decayed to 1×10^{-6} via cosine annealing. Early stopping based on validation loss is applied. The model is trained for 100 epochs on CAVE and 300 epochs on NTIRE2020/2022 to fully exploit its representation capacity.

To ensure an objective and comprehensive evaluation, all comparative models were implemented in PyTorch and trained on a single NVIDIA A40 GPU. Performance was assessed using seven widely adopted metrics: MRAE, root-mean-square error (RMSE), peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), spectral angle mapper (SAM), erreur relative globale adimensionnelle de synthèse (ERGAS), and correlation coefficient (CC).

The comparison results on the CAVE dataset are illustrated in Fig. 6. As shown in Fig. 6(a), it outperforms most existing methods in PSNR while maintaining a moderate floating-point operations (FLOPs) level and a compact model size. In Fig. 6(b), DiLAST demonstrates a relatively low and acceptable inference time. Compared to heavy models such as HRNet and DRCR, DiLAST achieves competitive or better reconstruction quality with significantly reduced computational overhead.

C. Ablation Studies

In this section, we conduct comprehensive ablation studies to investigate the effectiveness of the key components in the proposed DiLAST architecture. To ensure consistency and comparability, all experiments are carried out on the CAVE and DFC2018 Houston datasets for both training and evaluation. The primary objective is to assess the individual impact and sensitivity of each module on the overall performance. Furthermore, we present a systematic analysis and in-depth discussion of the contribution of each component to the final architecture.

1) Reconstruction With Spectral Differencing: To enhance spectral discriminability, we introduce a spectral differencing module that approximates the first-order spectral gradient via finite differences across RGB channels. Such gradients reflect key material transitions, serving as strong indicators of subtle spectral patterns. As illustrated in Fig. 7, while the original RGB channels exhibit limited variation, our operation amplifies interchannel differences, yielding more informative and discriminative features. Ablation results further validate the effectiveness of the proposed module. As shown in Table II, it achieves the best performance across most metrics, including a 0.271-dB PSNR gain and a notable ERGAS reduction compared to raw RGB input. Although Gaussian filtering slightly reduces SAM, our method delivers the overall best tradeoff, with the highest SSIM, demonstrating improved spectral fidelity and spatial consistency. These results underscore the value of incorporating spectral gradient priors to enhance feature sensitivity and reconstruction quality.

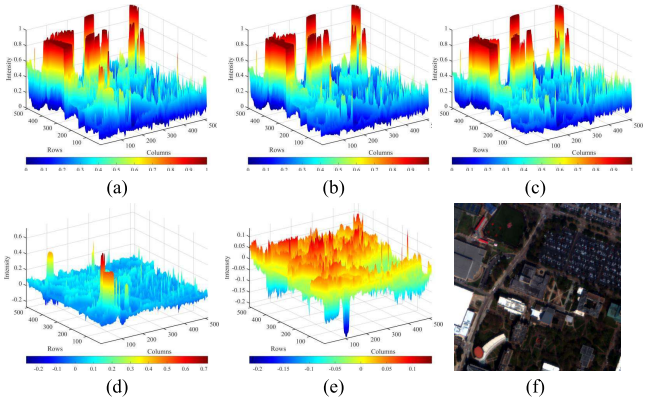


Fig. 7. Visualization of RGB channel distributions and spectral differences. (a) Red channel. (b) Green channel. (c) Blue channel. (d) (Red–green) channel. (e) (Blue–green) channel. (f) RGB image.

TABLE III
ABLATION STUDY OF DIFFERENT MSA SCHEMES

MSA Scheme	RMSE	PSNR	SSIM	SAM
Spe-MSA [9]	5.0054	35.603	0.9839	5.9384
Spa-MSA [65]	5.5208	34.073	0.9802	6.8267
Shift-Window MSA [57]	9.5358	30.803	0.9524	8.1741
Shuffle-Window MSA [66]	5.4952	34.313	0.9813	7.0105
GSSA	5.3609	34.595	0.9827	6.9461
CWSA	5.4068	34.474	0.9812	6.8309
CSSA (Ours)	**4.7325**	**36.026**	**0.9849**	**5.7913**

2) Validation of CSSA Effectiveness: We have evaluated CSSA against representative MSA variants, including Spectral-MSA [9], Spatial-MSA [65], Shifted Window MSA [57], Shuffle-Window MSA [66], GSSA, and CWSA. All variants were integrated into a unified model framework, with only the attention module replaced to ensure fair comparisons. Spatial-MSA focuses solely on spatial dependencies and incurs a high computational cost of $\mathcal{O}((HW)^2C)$. Localized variants, such as Shift-Window MSA and Shuffle-Window MSA, reduce the cost to $\mathcal{O}((HW)k^2C)$ by restricting attention to $k \times k$ windows while enhancing regional modeling capability through window shifting or shuffling. In contrast, Spe-MSA emphasizes spectral modeling, achieving a lower complexity of $\mathcal{O}(HWC^2)$. However, methods dedicated to either spatial or spectral attention cannot simultaneously capture long-range

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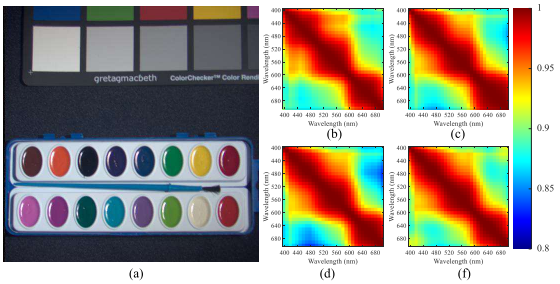


Fig. 8. We performed a visualization analysis of the spectral CCs for HSI reconstructed using different skip connection methods. (a) RGB image. (b) Ground truth. (c) Cat. (d) Add. (e) adaptive cross-scale fusion (ACF).

TABLE IV
ABLATION STUDY OF DIFFERENT SKIP CONNECTION STRATEGIES

Skip Connection Strategy	RMSE	PSNR	SSIM	SAM
Concatenation	5.7419	34.037	0.9806	7.3127
Addition	5.0178	35.189	0.9832	7.0304
ACF (Our)	**4.7325**	**36.026**	**0.9849**	**5.7913**

spatial patterns and global spectral correlations. To address this, we propose the CSSA, which decomposes attention into two efficient branches. The spatial branch (CWSA) adopts a center-window scheme to preserve local structures at a reduced cost of $\mathcal{O}((HW)k^2C)$, while the spectral branch (GSSA) applies global channelwise attention with a complexity of $\mathcal{O}(HWC^2)$ to model interband correlations. The overall complexity is, thus, $\mathcal{O}(HW(k^2C + C^2))$. This dual-branch design enables learning a structured joint attention matrix that approximates a low-rank decomposition of the full 3-D attention space, balancing performance and efficiency. As shown in Table III, CSSA consistently surpasses all counterparts across evaluation metrics, demonstrating its effectiveness in modeling both long-range spectral dependencies and nonlocal spatial patterns.

3) Influence of the ACF Module: Due to substantial differences in spatial structure and semantic abstraction, shallow and deep features are often misaligned. As shown in Table IV, simple addition or concatenation struggles to resolve this issue effectively. Elementwise addition averages feature responses but tends to suppress discriminative cues when fusing heterogeneous representations. Concatenation partially alleviates this by preserving channel diversity, yet it introduces parameter redundancy and suffers from limited reweighting capability. In contrast, the ACF module adaptively aligns multilevel features via dynamic channel attention, resulting in superior performance across all metrics. Its key advantages include: 1) the use of shared weights for unified projection of encoder and decoder features, which reduces redundancy and encourages focus on salient features and 2) a gating mechanism that explicitly models feature importance [see (15)], effectively suppressing misaligned information and enhancing semantic consistency across scales. As illustrated in Fig. 8, the similarity matrix of the HSI reconstructed using ACF exhibits a high degree of structural similarity with the ground truth, confirming its superiority in preserving fine-grained spatial details and high-level semantic features.

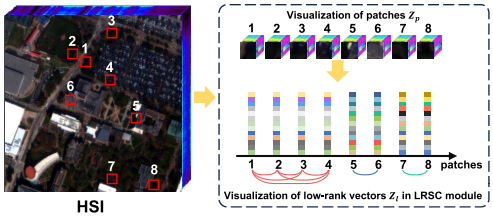


Fig. 9. Low-rank representation of spectral patches in the LRSC module.

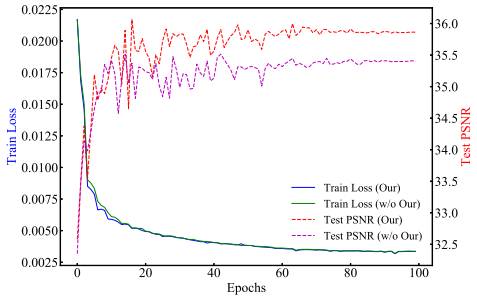


Fig. 10. Visualization of performance differences with and without the loss functions.

TABLE V
COMPARISON OF PERFORMANCE WITH AND WITHOUT PROPOSED LOSS FUNCTIONS

Loss Function	$\mathcal{L}_{MRAE}$	$\mathcal{L}_{sc}$	$\mathcal{L}_{ss}$	RMSE	PSNR	SSIM	SAM
$\mathcal{L}_{MRAE}$	✓	✗	✗	4.957	35.59	0.984	6.04
$\mathcal{L}_{sc}$	✓	✓	✗	4.919	35.73	0.984	6.12
$\mathcal{L}_{ss}$	✓	✗	✓	4.934	35.78	0.984	6.10
Our	✓	✓	✓	**4.733**	**36.03**	**0.985**	**5.79**

4) Analysis of the LRSC Module: We conduct a visual analysis of the low-rank vectors produced by the LRSC module to effectively demonstrate its effectiveness in spectral feature extraction, as illustrated in Fig. 9. In the DFC2018 Houston dataset, the original HSI data provide an initial visual representation of the similarities among ground objects. However, certain land-cover types present challenges in visual differentiation. For instance, in patches 1–4, it remains visually ambiguous whether these regions belong to the same class. In contrast, after processing with the LRSC module, the resulting low-rank vectors exhibit a high degree of consistency, significantly outperforming those obtained from other regions. These results indicate that the LRSC module is adept at uncovering latent relationships among image patches and adaptively extracting discriminative spectral features, thereby enhancing object separability in HSI.

5) Role of the Loss Function: We conducted a comparative experiment to evaluate the effectiveness of incorporating the proposed spectral consistency loss $\mathcal{L}_{\text{sc}}$ and spatial smoothness loss $\mathcal{L}_{\text{ss}}$ during training. As shown in Table V, introducing either loss individually leads to a certain degree of improvement in reconstruction performance. However, when both losses are used simultaneously, the model achieves the best results across multiple evaluation metrics. To further analyze the training dynamics, we visualized the loss values and the corresponding PSNR trends throughout the training process

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TABLE VI

COMPARATIVE ANALYSIS OF THE IMPACT OF DIFFERENT MODULES ON PERFORMANCE

Configuration	CSSA	ACF	LRSC	RMSE	PSNR	SSIM	SAM
Baseline	✗	✗	✗	5.7623	33.8861	0.9799	6.5774
w/ CSSA	✓	✗	✗	5.0727	35.4406	0.9840	6.0834
w/ ACF	✗	✓	✗	5.4393	34.2434	0.9807	6.5647
w/ LRSC	✗	✗	✓	5.7459	33.8985	0.9799	6.5785
w/o CSSA	✗	✓	✓	5.5208	34.0734	0.9802	6.8267
w/o ACF	✓	✗	✓	5.7419	34.0365	0.9806	7.3127
w/o LRSC	✓	✓	✗	4.9570	35.5904	0.9840	6.0400
DiLAST	✓	✓	✓	**4.7330**	**36.0255**	**0.9850**	**5.7910**

TABLE VII

PERFORMANCE COMPARISON WITH SOTA METHODS ON THE CAVE DATASETS. BEST RESULTS ARE IN BOLD, AND THE SECOND-BEST ARE UNDERLINED

Method	MRAE	RMSE	PSNR	SSIM	SAM	ERGAS	CC
HRNet	0.0098	5.1359	35.028	0.9810	6.7772	18.831	0.9939
MIRNet	0.0099	5.3931	35.179	0.9813	6.4876	19.256	0.9932
HINet	0.0103	5.2608	34.847	0.9739	7.4425	20.007	0.9934
MST++	0.0091	4.9895	35.201	0.9837	6.2759	18.552	0.9945
DRCR	0.0097	5.4444	34.593	0.9790	7.3352	20.119	0.9928
HDNet	0.0105	5.6529	34.413	0.9801	6.7067	20.462	0.9928
RepCPSI	0.0097	5.2418	34.942	0.9816	6.5291	18.816	0.9940
HySAT	0.0097	5.2615	34.995	0.9825	6.7282	19.502	0.9941
GMSR	0.0105	5.5273	34.488	0.9772	7.3202	19.806	0.9927
CESST	0.0095	5.2669	34.766	0.9827	6.6505	18.586	0.9946
Ours	**0.0086**	**4.7325**	**36.026**	**0.9849**	**5.7913**	**18.071**	**0.9948**

(see Fig. 10). The results indicate that the inclusion of the proposed loss functions significantly accelerates the model's convergence and enables it to reach peak PSNR performance earlier on the test set.

6) Effectiveness of Individual Modules: Table VI reports ablation results of the three modules in DiLAST: CSSA, ACF, and LRSC. CSSA captures spatial similarity and spectral correlation via a dual-branch attention scheme, yielding a 1.5545-dB PSNR gain and notable RMSE reduction. ACF adaptively fuses multiscale features through channelwise attention, improving spectral fidelity and reducing SAM. LRSC enforces low-rank spectral regularization guided by global memory, enhancing spatial–spectral coherence and stability. The joint integration yields additional performance gains; for instance, the full DiLAST achieves a 2.1394-dB improvement in PSNR and an approximately 1% reduction in RMSE, confirming that these improvements result from complementary and physically motivated designs rather than increased model complexity.

D. Comparisons With Other Methods

We conduct a comparative analysis of the proposed DiLAST framework against several SOTA methods, including HRNet [23], MIRNet [67], HINet [41], MST++ [9], DRCR [32], HDNet [51], RepCPSI [50], HySAT [25], GMSR [34], and CESST [10]. We evaluate all methods using quantitative and qualitative metrics to assess robustness and generalization. Baselines follow official settings, with the best-performing versions tested on independent datasets.

TABLE VIII

PERFORMANCE COMPARISON WITH SOTA METHODS ON THE NTIRE2020 DATASETS. BEST RESULTS ARE IN BOLD, AND THE SECOND-BEST ARE UNDERLINED

Method	MRAE	RMSE	PSNR	SSIM	SAM	ERGAS	CC
HRNet	0.1319	0.0342	32.754	0.9698	3.1719	15.924	0.9810
MIRNet	0.0831	0.0184	36.353	0.9883	3.0164	8.7187	0.9962
HINet	0.3861	0.0902	21.759	0.8558	3.3019	44.159	0.9947
MST++	0.0977	0.0174	36.489	0.9863	2.9631	8.4441	0.9976
DRCR	0.0832	0.0186	36.369	0.9861	2.8676	8.7440	0.9974
HDNet	0.0926	0.0178	36.355	0.9855	3.0483	8.7942	0.9964
RepCPSI	**0.0729**	0.0171	36.993	0.9873	3.0236	**7.8202**	0.9960
HySAT	0.0883	0.0177	36.503	0.9857	2.8953	9.0233	0.9976
GMSR	0.2106	0.0485	27.103	0.9594	3.4639	22.169	0.9942
CESST	0.1208	0.0200	35.249	0.9808	3.0565	10.558	0.9971
Ours	0.0817	**0.0169**	**37.014**	**0.9900**	**2.4187**	7.9998	**0.9977**

TABLE IX

PERFORMANCE COMPARISON WITH SOTA METHODS ON THE NTIRE2022 DATASETS. BEST RESULTS ARE IN BOLD, AND THE SECOND-BEST ARE UNDERLINED

Method	MRAE	RMSE	PSNR	SSIM	SAM	ERGAS	CC
HRNet	0.3722	0.0539	27.039	0.8683	5.3757	67.563	0.9074
MIRNet	0.2190	0.0295	32.282	0.9448	4.8070	35.708	0.9860
HINet	0.2392	0.0317	31.891	0.9391	5.0118	43.586	0.9803
MST++	**0.1824**	**0.0261**	33.567	0.9513	4.9051	**32.744**	0.9856
DRCR	0.3043	0.0355	31.242	0.9474	4.6309	34.838	0.9865
HDNet	0.2785	0.0383	29.811	0.9058	5.6710	47.641	0.9654
RepCPSI	0.2145	0.0291	32.415	0.9435	5.1230	35.055	0.9831
HySAT	0.2013	0.0290	32.951	0.9417	4.7229	36.878	**0.9870**
GMSR	0.4088	0.0687	26.019	0.8079	6.2218	63.002	0.9621
CESST	0.1899	0.0267	33.789	0.9519	4.8514	33.274	0.9851
Ours	0.1836	0.0262	**33.820**	**0.9524**	**4.6029**	32.965	0.9868

1) Quantitative Results: As shown in Tables VII–IX, we conducted quantitative evaluations on three benchmark datasets: CAVE, NTIRE2020, and NTIRE2022. DiLAST achieves the best performance on most metrics for CAVE and NTIRE2020, demonstrating superior reconstruction quality. On the more challenging NTIRE2022 dataset, DiLAST ranks first in SSIM, PSNR, and SAM, but slightly behind in MRAE and CC. This is mainly due to the increased complexity of the dataset, including more diverse materials, which makes accurate spectral recovery more difficult. Nevertheless, the results across multiple datasets also confirm the robustness of our model.

2) Qualitative Evaluation: To further validate the superior performance of DiLAST, we conducted qualitative analyses on representative samples from CAVE, NTIRE2020, and NTIRE2022-Specifically, "Fake Lemon/Real Lemon," "ARAD_HS_0465_clean," and "ARAD_1K_0949." Figs. 11–13 present RMSE heatmaps at the 600- and 690-nm spectral bands for various comparative methods, providing intuitive visual assessments of spatial reconstruction quality. In the heatmaps, darker regions indicate lower reconstruction errors, while brighter areas correspond to higher deviations. DiLAST consistently yields RMSE distributions that align more closely with the ground truth, outperforming representative SOTA methods such as MST++, GMSR, and CESST across all test cases, consistent with prior quantitative results. To assess spectral fidelity, we further visualized spectral

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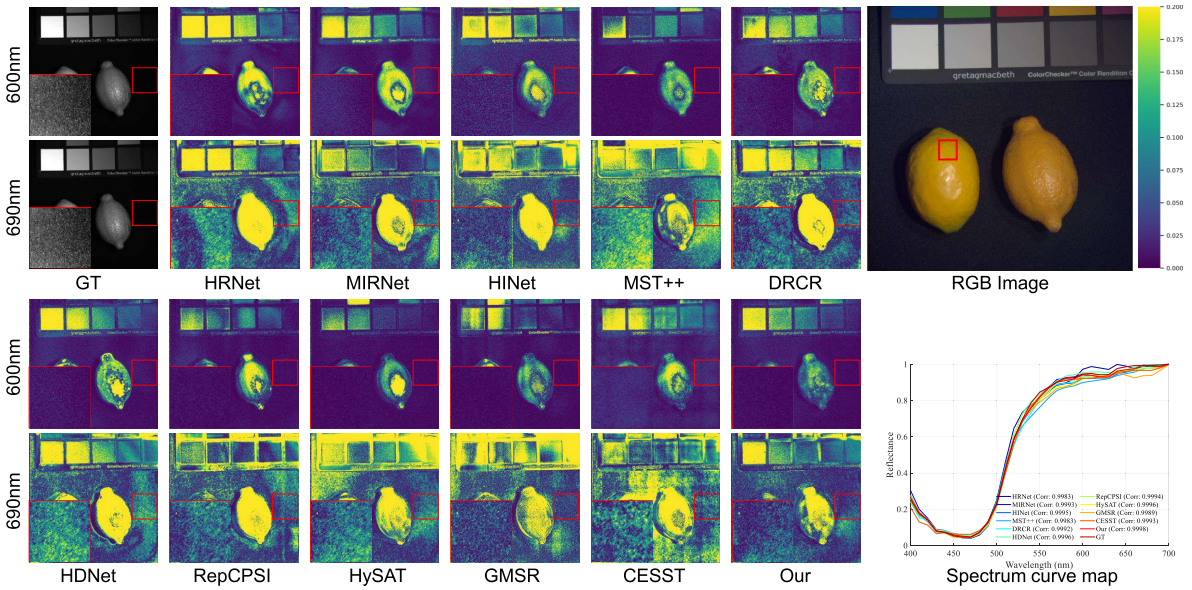


Fig. 11. Qualitative visualization on the CAVE dataset: RMSE maps at 600 and 690 nm are shown for each SSR method and the proposed method. The spectral curves on the right correspond to the red-boxed region in the RGB image. Please zoom in for a clearer visual comparison.

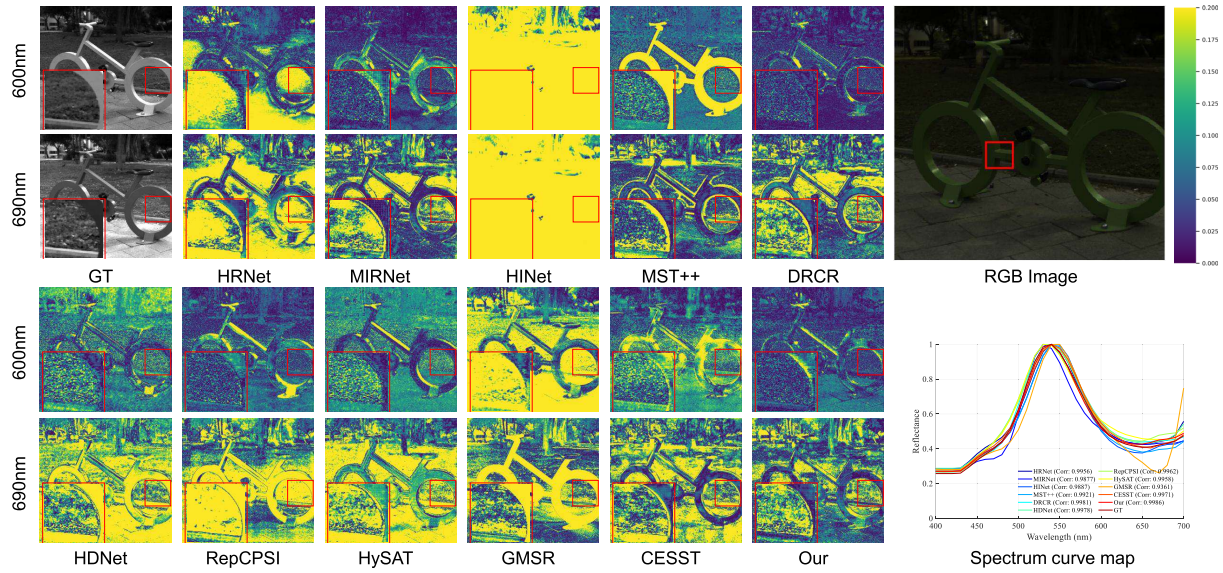


Fig. 12. Qualitative visualization on the NTIRE2020 dataset: RMSE maps at 600 and 690 nm are shown for each SSR method and the proposed method. The spectral curves on the right correspond to the red-boxed region in the RGB image. Please zoom in for a clearer visual comparison.

reflectance curves from the red-marked regions. Since the shape of spectral curves directly reflects material properties, their accurate reconstruction serves as a key indicator of spectral quality. DiLAST closely matches the ground truth in both shape and trend, demonstrating strong local consistency. Overall, spatial and spectral analyses validate the robustness of DiLAST and its effectiveness in preserving spectral continuity while modeling multiscale spatial–spectral dependencies.

E. Comparison of Remote Sensing Scene Reconstruction Performance

To assess real-world robustness, additional experiments were conducted on a representative remote sensing dataset with complex backgrounds. Specifically, we employed the DFC2018 Houston dataset, which was collected by the National Center for Airborne Laser Mapping, University of Houston. This dataset has a spatial resolution of 4172 × 1202 pixels and consists of 48 spectral bands covering the wavelength range of 380–1050 nm. Following the methodology of Liu et al. [68], we selected bands 23, 12, and 5 as the RGB channels for visualization and divided the original image into 27 patches of size 512 × 512, with three patches used for testing and the remaining ones for training.

We compare it with two representative SOTA approaches, MST++ and RepCPSI, both of which have demonstrated strong performance in previous experiments. Table X reports the quantitative results of all methods on the DFC2018 Houston dataset. Due to the diverse land-cover types and complex

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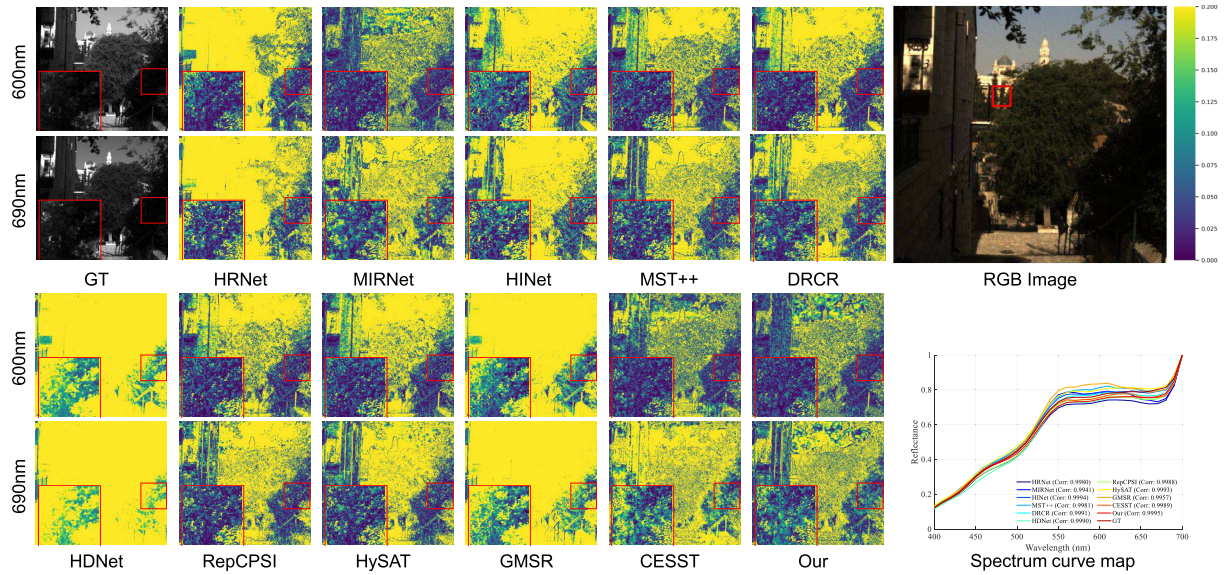


Fig. 13. Qualitative visualization on the NTIRE2022 dataset: RMSE maps at 600 and 690 nm are shown for each SSR method and the proposed method. The spectral curves on the right correspond to the red-boxed region in the RGB image. Please zoom in for a clearer visual comparison.

TABLE X
COMPARISON OF EVALUATION METRICS FOR ADVANCED SSR METHODS ON THE HOUSTONU 2018 DATASET

Method	MRAE	RMSE	PSNR	SSIM	SAM
RepCPSI	0.1707	66.0219	14.9963	0.6049	10.2250
MST++	0.1687	64.9632	15.2152	0.6066	10.0304
DiLAST (Our)	**0.1671**	**63.2520**	**15.4030**	**0.6002**	**9.8434**

object boundaries in remote sensing imagery, all methods exhibit a slight performance degradation compared to that on natural image datasets. Nevertheless, the proposed method achieves the best performance across all five evaluation metrics, indicating superior generalization ability and robustness.

Since RGB images are constructed by selecting specific bands from hyperspectral data, we also visualize the reconstructed results of each method at bands 23, 12, and 5 as pseudo-color images to evaluate their ability to recover key spectral channels. As shown in Fig. 14, although the proposed method demonstrates clear advantages in terms of quantitative performance, noticeable visual differences remain compared to the ground truth. This highlights the inherent complexity of remote sensing scenes, which must be further addressed to improve the perceptual quality and practical applicability of HSI reconstruction. Compared to natural scenes in the CAVE dataset, remote sensing data pose additional challenges, including illumination variations from sunlight and shadows, diverse materials with complex spectral signatures, mixed pixels by limited spatial resolution, and sensor or atmospheric noise. These factors contribute to lower PSNR and SSIM scores despite visually plausible reconstructions.

F. Improvement in Downstream Task Performance

In real-world scenarios, the lack of precisely aligned HSI corresponding to RGB inputs poses a significant challenge

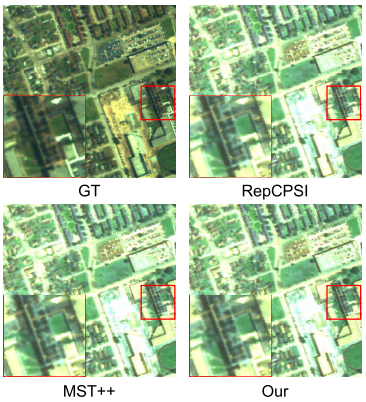
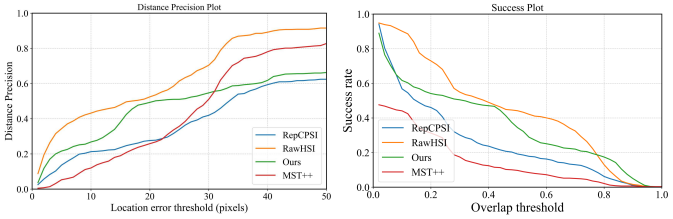


Fig. 14. False-color visualizations of DFC2018 Houston HSI synthesized by different methods (bands 23, 12, and 5). Please zoom in for a clearer visual comparison.

for traditional quantitative evaluation, making it difficult to directly assess the performance of SSR algorithms. In addition, the inherent complexity of HSI along the spectral dimension limits the effectiveness of intuitive visual comparisons across different reconstruction methods. HSI offers strong pixel-level discriminative capability, which is particularly useful for high-level vision tasks such as object tracking. We conduct an object tracking experiment on hyperspectral video (HSV) as a complementary downstream task. Specifically, a tracking framework is employed to indirectly evaluate the impact of different SSR methods in scenarios without ground-truth spectral labels.

1) Experimental Settings: We employ the VIS dataset from the Whisper 2024 hyperspectral object tracking challenge for downstream task evaluation. The dataset was captured using an XIMEA snapshot VIS camera and encompasses a variety of real-world scenes. Four video sequences were randomly selected: one for training (550 frames), one for validation (450 frames), and two for testing (270 and 437 frames,

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(a) (b)

Fig. 15. Comparison of different methods evaluated with the visual tracking task. (a) Distance plot. (b) Success plot.

TABLE XI
COMPARISON OF VISUAL TRACKING PERFORMANCE ACROSS METHODS ON HYPERSPECTRAL DATA

Method	AUC ↑	DP@20 ↑
RawHSI	0.4365	0.5242
RepCPSI	0.2511	0.2761
MST++	0.1475	0.2583
DiLAST (Our)	**0.3522**	**0.4929**

respectively). Each scene contains both HSV and the corresponding RGB sequences, with a spatial resolution of 512 × 272 pixels per frame. The spectral range spans from 470 to 620 nm, comprising 16 spectral bands with a uniform interval of 10 nm.

MST++ and RepCPSI were selected as baseline methods for comparison, as both have demonstrated competitive yet suboptimal performance in previous experiments. During training, each video frame was divided into 18 partially overlapping patches of size 128 × 128 to construct the training samples. Due to the limited variation between consecutive frames, directly training on full-frame sequences can hinder model generalization. To address this, we applied data augmentation techniques by randomly flipping and rotating each training patch with a 50% probability, thereby increasing sample diversity and mitigating overfitting.

During testing, the reconstructed HSV was fed into SSTtrack [2], a pretrained tracker designed for HSV. Tracking performance using the original HSI (denoted as "RawHSI") was treated as the upper bound for each reconstruction method. Evaluation was conducted using average precision (AP) and area under the curve (AUC).

2) Quantitative Results: Fig. 15 and Table XI present the experimental results on the object tracking task. The DiLAST method not only significantly outperforms MST++ and RepCPSI in tracking performance, but also closely approaches the upper bound results achieved under the "RawHSI" setting. As expected, all computational SSR methods yield lower tracking performance compared to the "RawHSI" baseline using ground-truth HSI, primarily due to inevitable reconstruction errors. Notably, DiLAST exhibits a clear advantage in narrowing this performance gap relative to MST++ and RepCPSI. These results confirm that the proposed center spatial–spectral modeling strategy and LRSC mechanism more effectively capture and preserve critical spectral information, thereby maintaining spectral fidelity while enhancing the discriminative power of the reconstructed data for high-level vision tasks. In summary, DiLAST achieves superior results in spectral reconstruction metrics and demonstrates strong applicability to downstream tasks, validating its effectiveness and robustness as a practical SSR solution.

V. DISCUSSION

Despite the strong performance of DiLAST across multiple datasets, certain limitations remain in complex scenarios. First, on the DFC2018 Houston dataset, the model exhibits performance degradation under challenging remote sensing conditions such as extreme illumination variations (e.g., shadows and specular reflections) and high spectral similarity between materials. In such cases, different land-cover classes may appear visually similar in the RGB domain but differ significantly in the hyperspectral domain, making it difficult for the model to recover fine-grained spectral information. This issue becomes more pronounced in mixed-pixel regions or under low signal-to-noise conditions, highlighting the increased difficulty of spectral reconstruction in remote sensing scenes compared to natural indoor datasets. Second, the current model is trained on clean, high-quality RGB images and does not explicitly consider common degradations such as compression artifacts or sensor-induced noise. As a result, the reconstruction accuracy may deteriorate when dealing with degraded inputs. Although the spectral consistency modeling within DiLAST provides a certain level of robustness, there is still room for improvement. Future work may explore noise-aware training, joint denoising-reconstruction frameworks, or compression-invariant feature extraction to enhance the model's generalization ability under real-world degraded conditions.

VI. CONCLUSION

In this article, we propose a novel framework, termed DiLAST, for HSI-SSR from RGB images. By incorporating cross-domain spatial–spectral modeling, DiLAST jointly captures nonlocal spatial dependencies, global spectral correlations, and low-rank spectral priors in a coarse-to-fine manner. The framework comprises three key components: the CSSA module for enhancing joint spatial–spectral features, the ACF module for dynamically aligning hierarchical representations, and the LRSC module that exploits spectral redundancy via a learnable memory mechanism. Extensive experiments on the benchmark datasets show that DiLAST consistently outperforms existing SSR methods, achieving superior spectral fidelity and spatial consistency. Moreover, downstream evaluations on remote sensing image reconstruction and object tracking further demonstrate its effectiveness and robustness. These results highlight the strong generalization capability of DiLAST, particularly in remote sensing applications.

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